

Moshiur Rahman Faisal

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Dhaka - 1229, Bangladesh

OBJECTIVE

I'm a motivated AI researcher with hands-on experience in the telecommunications sector, where I've worked on AI-driven, data-focused solutions to tackle real-world challenges. I'm skilled in both qualitative and quantitative research, with a background in data analysis, natural language processing (NLP), and building machine learning (ML) systems. My passion lies in pushing the boundaries of AI, especially in NLP, and I'm aiming to pursue a Ph.D. to deepen my expertise. I enjoy working across disciplines and collaborating with diverse teams to drive meaningful innovations in AI research.

EDUCATION

• North South University

January 2018 - April 2022

Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

◦ CGPA: 3.59/4.00 (Above 89% according to NSU Grading Policy)

◦ Award: Cum Laude

◦ Senior Design Project (Thesis): Text Based Emotion Detection (Banglish, Bangla & English)

Supervisor: Associate Prof. Dr. Md Shahriar Karim

Summary: Developed a low-resource emotional corpus for transliterated Bengali (Banglish) text from social media, using CNN + BiLSTM (FastText) and Random Forest (Tf-Idf) models for emotion classification, enhancing sentiment analysis in Bengali language processing.

◦ Directed Research: Mental Instability Detection From Text Data

Supervisor: Prof. Dr. Sifat Momen

Summary: Developed a text-based mental health prediction system using machine learning, deep learning, and transformer models to detect psychological conditions from social media text in both Bangla and English, enhancing early detection and intervention for mental health issues.

RESEARCH EXPERIENCE

• North South University

Nov 2023 - Dec 2024

Senior Research Assistant (Part-time)

Remote

Supervisor: Prof. Dr. Mohammad Rashedur Rahman

◦ Led a team of researchers in conducting advanced natural language processing (NLP) studies focused on Bengali and Banglish text, improving model accuracy and performance for contextual text classification.

◦ Spearheaded the design and implementation of a context-aware data cleaning pipeline to optimize Bengali text, resulting in improved performance for text classification models.

◦ Developed a comprehensive monolingual dataset for emotion detection in Bengali and Banglish, enabling more nuanced analysis and improving model robustness in sentiment analysis tasks.

◦ Applied cutting-edge machine learning and NLP techniques to enhance dataset quality, guide team efforts, and ensure the successful completion of research goals.

INDUSTRY EXPERIENCE

• Robi Axiata PLC.

Nov 2022 - Present

Senior Associate, AI & Data Science

Dhaka, Bangladesh

◦ Collaborated with cross-functional teams to define requirements and design scalable Big Data pipelines using Spark and Hadoop for efficient analytics.

◦ Developed and deployed AI/ML models, including Large Language Models (LLMs), leveraging AWS to solve business challenges and enhance operational efficiency.

◦ Conducted applied AI research to evaluate novel algorithms and bridge state-of-the-art methods with practical business solutions.

◦ Created interactive dashboards in Tableau and integrated AI-driven solutions into processes, improving automation, customer engagement, and decision-making.

• AlterSense Limited

May 2022 - Oct 2022

Junior Machine Learning Engineer

Dhaka, Bangladesh

◦ Processed and managed large-scale image and video datasets to train and optimize deep learning models, including YOLOv5, for advanced object detection and analysis.

◦ Conducted applied research to develop, train, and deploy deep learning models using TensorFlow and OpenCV, integrating them into production pipelines with Docker for continuous operation.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL

- [C.1] Maheen, Syed Mustavi, Faisal, Moshir Rahman, Rahman, Md. Rafakat, & Karim, Md. Shahriar (2022, July). **Alternative non-BERT model choices for the textual classification in low-resource languages and environments**. *Proceedings of the Third Workshop on Deep Learning for Low-Resource Natural Language Processing*, pp. 192–202. Association for Computational Linguistics. Hybrid. DOI: 10.18653/v1/2022.deeplo-1.20
- [J.1] Faisal, Moshir Rahman, Shifa, Ashrin Mobashira, Rahman, Md. Hasibur, Uddin, Mohammed Arif, & Rahman, Rashedur M. (2024). **Bengali & Banglish: A monolingual dataset for emotion detection in linguistically diverse contexts**. *Data in Brief*, 55, 110760, pp. 1–10. DOI: 10.1016/j.dib.2024.110760
- [J.2] Faisal, Moshir Rahman, Fahad, Abdur Rahman, Ridoy, Shahriyar Zaman, Sultana, Jannat, Ria, Zinnat Fowzia, Rahman, Md. Hasibur, Uddin, Mohammed Arif, & Rahman, Rashedur M. (2025). **Context-Aware Data Cleaning: Optimizing Bengali Text for Contextual Text Classification**. *SN Computer Science*, 6(5), 422. DOI: 10.1007/s42979-025-03891-9

PROJECTS

- **Telco Churn Prediction and Explainability with AI** Apr 2025 - July 2025
Tools: [Python, Pandas, Scikit-learn, Matplotlib, Seaborn, SHAP, LIME, AWS]
 - Developed machine learning models to predict customer churn using telco data, performing data preprocessing and leveraging SHAP and LIME for interpretability to identify key churn factors.
- **Customer Insight 360** Apr 2024 - Nov 2024
Tools: [Python, Apache Airflow, Apache Spark, Oracle SQL, PL/SQL, Docker, Linux, Tableau]
 - Developed data pipelines to process data from over 51 million customers daily using Hadoop and Oracle. Leveraged machine learning for customer predictions and tracked 250+ KPIs, visualized in Tableau dashboards.
- **RMG Safety and Activity Monitoring** Jun 2022 - Oct 2022
Tools: [Python, Darknet, PyTorch, Docker, OpenCV, Linux]
 - Processed live CCTV footage for deep learning model training. The best model was deployed using Docker and automated data pipelines for real-time activity monitoring and reporting to business users.

SKILLS

- **Programming Languages:** Python, C, C++, Java, R
- **Database Systems:** Oracle, Apache Hive, MySQL
- **Data Science & Machine Learning:** TensorFlow, Keras, PyTorch, Scikit-learn, Hugging Face Transformers, Ktrain, Pandas, OpenCV, NLTK, Matplotlib, Seaborn, Apache Spark
- **Cloud Technologies:** Cloudera, AWS, Google Colab
- **DevOps & Version Control:** Apache Airflow, Docker, Hadoop, Git, GitLab CI/CD, Linux
- **Specialized Area:** Natural Language Processing (NLP), Deep Learning, Big Data, Data Engineering, Docker-based Development, Database Process Automation
- **Other Tools & Technologies:** Tableau, Power BI, Jupyter Notebooks, Microsoft 365, Trello
- **Research Skills:** Quick Learner, Problem Analysis, Documentation, Time Management, Teamwork, Collaboration, Overleaf

HONORS AND AWARDS

- **Q2 2024 Best Team Efforts Award** Aug 2024
Corporate Strategy, Robi Axiata Limited
 - Awarded for planning and leading a major nationwide AI-focused datathon, fostering innovation, collaboration, and data-driven problem-solving across the country.
- **Q2 2024 Star Performer Award** Aug 2024
Corporate Strategy, Robi Axiata Limited
 - Awarded for designing and implementing an efficient AI-driven data processing system, capable of handling over 50 million records daily.
- **Q1 2023 Star Performer Award** Jun 2023
Corporate Strategy, Robi Axiata Limited
 - Awarded for developing a comprehensive weather data pipeline and predictive model, integrated with a Tableau dashboard, to analyze the impact of weather conditions on business performance in specific regions.
- **NAACL Diversity & Inclusion Subsidy Award** Jun 2022
2022 Annual Conference of the North American Chapter of the Association for Computational Linguistics
 - Awarded the Diversity & Inclusion Subsidy for demonstrating a strong commitment to fostering diversity in the field of computational linguistics.

VOLUNTEER EXPERIENCE

- **Chairperson**

IEEE North South University (NSU) Student Branch

◦ Led the IEEE NSU Student Branch, overseeing a team of student volunteers and coordinating branch activities, events, and workshops.
- **Country Lead**

IEEEExtreme 15.0

◦ Led the coordination and execution of IEEEExtreme 15.0, an international programming competition, across the country, involving more than 45 universities and 135 student participants.
- **Graphic Design Assistant**

IEEE DAY 2021

◦ Led the graphics team in designing promotional materials, event banners, and social media graphics, improving the visibility and branding of IEEE Day 2021.

CERTIFICATIONS

- **ETL and Data Pipelines with Shell, Airflow and Kafka**

Nov 2024
- **Inferential Statistics**

Oct 2024
- **Machine Learning with Python**

Sep 2024
- **Generative AI with Large Language Models**

Mar 2024
- **SQL (Advanced) Certificate**

Mar 2024
- **Machine Learning Specialization**

Jan 2024
- **Python for Data Science, AI & Development**

Jan 2024

ADDITIONAL INFORMATION

Languages: English (IELTS 7.0), Bengali (Native)
Interests: Natural Language Processing (NLP), Machine Learning Operations (MLOps), Data-Driven Product Development, Technology Innovations

REFERENCES

1. **Dr. Md Shahriar Karim**

Associate Professor, Department of Electrical & Computer Engineering

North South University

Email: shahriar.karim@northsouth.edu

Relationship: Thesis Supervisor
2. **Dr. Sifat Momen**

Professor, Department of Electrical & Computer Engineering

North South University

Email: sifat.momen@northsouth.edu

Relationship: Directed Research Supervisor
3. **Dr. Mohammad Rashedur Rahman**

Professor, Department of Electrical & Computer Engineering

North South University

Email: rashedur.rahman@northsouth.edu

Relationship: Research Supervisor